Safety Immersion and Gamified Hazard Training (SIGHT)

Research Assistant Packet

VR session: web-based virtual reality on a 27-inch monitor · seated at a tabletop

Participant ID	**P** _____	**Session date**	_____
VR email	p_____@sight.study	**VR password**	on the login card
Machine order	Order ____ (see Overview #1)		
Research assistant	_____	**Start time**	_____

Every script and research assistant form for one VR session, in the order you use them. The participant's forms (consent and questionnaires) are a separate PDF; hand them over one at a time. Print both single-sided and do not staple, so the signed consent form can be filed separately.

Packet Contents

Overview

#1 Session at a Glance . . . 2

#2 Research Assistant Run Sheet . . . 3

Experiment

#1 Setup, Orientation & VR Login (10 min) . . . 5

#2 Observation Sheet: Machine 1 (15 min) . . . 6

#3 Observation Sheet: Machine 2 (15 min) . . . 7

#4 Observation Sheet: Machine 3 (15 min) . . . 8

End of Session

#1 Assessment Modules (about 10 min) . . . 9

#3 Interview (about 5–10 min) . . . 10

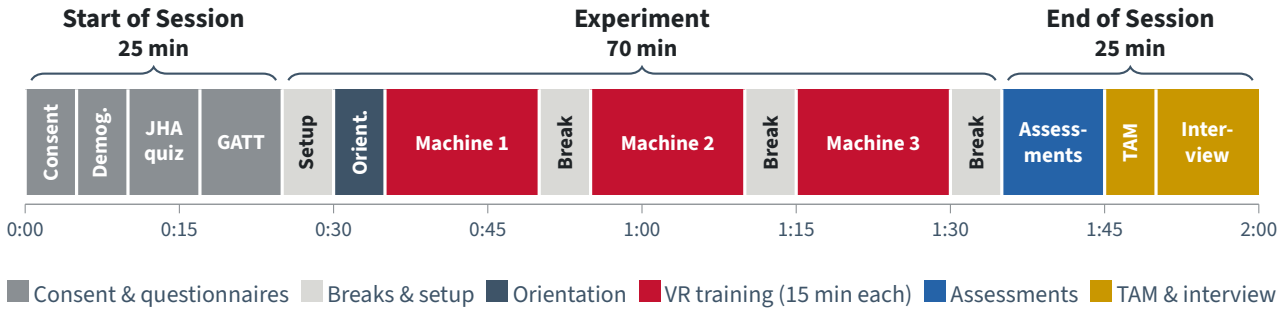
#4 Close-out & Data Entry . . . 12

P **Participant forms** (separate PDF): Start of Session #1–#4 and End of Session #2.

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Session at a Glance

Schematic of the VR session, with planned minutes for every step



Planned total **about 120 minutes**. Orientation (5 min) and assessments (10 min) are estimates.

This participant

Participant ID	**P** _____ (order _____)
Machines	_____ → _____ → _____
VR site	`sightvr.org`
VR email	`p_____@sight.study`
VR password	on the login card

Materials

- ☐ This packet and the participant forms, printed single-sided, unstapled
- ☐ Pens, clipboard, timer (15:00 per machine)
- ☐ Computer, 27" monitor, keyboard, mouse; chair at a tabletop
- ☐ Audio recorder, set to m4a or mp3 (only if audio consent applies)
- ☐ $75 Visa gift card for the participant
- ☐ The VR login card (keep it with you; never scan it)

Contacts

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Research Ethics & Integrity: (513) 529-3600, humansubjects@miamioh.edu

Machine orders (protocol draft)

Order	1st	2nd	3rd
1	Mill	CNC Lathe	Cobot
2	Mill	Cobot	CNC Lathe
3	CNC Lathe	Mill	Cobot
4	CNC Lathe	Cobot	Mill
5	Cobot	Mill	CNC Lathe
6	Cobot	CNC Lathe	Mill

Participant n uses order $((n-1) \bmod 6) + 1$: P1 and P7 use order 1, P2 and P8 order 2, and so on. A = Mill, B = CNC Lathe, C = Cobot.

Key rules

- Stop each machine at 15:00. Do not push the participant to finish.
- Give a 5-minute break after the baseline forms and after each machine.
- Fatigue, pain, dizziness, or loss of balance: you can pause and/or stop the experiment. Note it on the observation sheet.
- Hand over participant forms only; RA pages stay with you.
- Never write a participant's name on any page.

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Research Assistant Run Sheet

Keep this sheet beside you. Tick each step and write the clock time.

Before the participant arrives

- ☐ Check that every page of this packet and of the participant forms is present (see each cover).
- ☐ Set up a chair at the tabletop, the computer, 27" monitor, keyboard, and mouse.
- ☐ Have pens and the timer ready.

	Planned	Clock time	Step	Document
Start of Session				
☐	0:00		Welcome and set expectations (script below)	This sheet
☐	0:00		Consent form (about 5 min)	Start of Session #1 P
☐	0:05		Demographic questionnaire (about 5 min)	Start of Session #2 P
☐	0:10		Job Hazard Analysis quiz (about 7 min)	Start of Session #3 P
☐	0:17		GATT questionnaire (about 8 min)	Start of Session #4 P
Experiment				
☐	0:25		5-minute break; you set up the VR system	Experiment #1
☐	0:30		VR orientation and study login (about 5 min)	Experiment #1
☐	0:35		Machine 1: _____ (15 min, timed)	Experiment #2
☐	0:50		5-minute break	
☐	0:55		Machine 2: _____ (15 min, timed)	Experiment #3
☐	1:10		5-minute break	
☐	1:15		Machine 3: _____ (15 min, timed)	Experiment #4
☐	1:30		5-minute break	
End of Session				
☐	1:35		Assessment modules, one per machine, in training order (about 10 min)	End of Session #1
☐	1:45		TAM questionnaire (about 5 min)	End of Session #2 P
☐	1:50		Interview (about 10 min)	End of Session #3
☐	2:00		Close out ($75 Visa gift card); data entry within 24 hours	End of Session #4

P = in the participant forms PDF; every other document is in this packet.

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Start of Session scripts

Welcome and set expectations. Introduce yourself as a researcher and thank the participant for taking part. You and the participant stay seated for the rest of the session.

> **SAY** *Today, you are helping to improve the SIGHT safety training system by giving feedback about aspects such as ease of use, engagement, and effects on performance, while also comparing this experience to your prior experience with similar safety training. You will be given the system to try out. Then, you will be asked to complete a survey and be interviewed about your experience and opinions of the systems.*

Consent (Start of Session #1). Give the participant the consent form. Have them read it, or read it through with them, and answer any questions.

> **SAY** *Before we continue, I would like you to read and sign this consent form. This explains the research study and your role and rights throughout the process. I am happy to answer any questions you might have. I also have a copy of the consent form that I will leave with you.*

- ☐ The participant and you sign and date the form.
- ☐ Give the participant the second consent copy, marked *Your copy, please keep*.
- ☐ Take the signed form out of this packet right away and keep it apart from the participant's data; at data entry it is scanned into the **Signed Consent Forms** folder. **Never write the participant ID on the consent form**: the form promises that answers cannot be linked to names.

Baseline questionnaires (Start of Session #2–#4). Hand over one form at a time. You may answer questions about the forms, but do not help with quiz answers.

> **SAY** *First, please complete these baseline questionnaires. These will include a demographic questionnaire, a quiz on your current knowledge of job hazards, and a questionnaire about your attitudes toward technology. You may leave questions blank that you do not want to answer.*

Health question. The last question on the demographic form is the approved screening question (protocol 1.1). Check it before you hand over the next form. If the answer is YES, the person cannot take part: thank them and end the session. Give the $75 Visa gift card, as for any session that ends early.

Then give the participant a 5-minute break and go to Experiment #1.

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Setup, Orientation & VR Login

5-minute participant break, then about 5 minutes of orientation

During the 5-minute break, set up

- ☐ Chair at the tabletop (the participant stays seated); computer, 27" monitor, keyboard, mouse.
- ☐ Browser open at the SIGHT VR site: `sightvr.org`.
- ☐ Timer ready at 15:00. Observation sheets (Experiment #2–#4) at hand.

Machines, in order: 1. _____ → 2. _____ → 3. _____ (order ____)

Orientation. Give a quick overview of the VR setup: point out the module menu, where the on-screen instructions appear, and how to move, look, and select with the mouse and keyboard.

> **SAY** *This is the SIGHT VR safety training system. You will work through the training on each of 3 machines: lathe, mill, and collaborative robot. As you work through the training, if anything is confusing or difficult to use, try to work through it as you normally would if you bought the system yourself. We will discuss your experience later in the session.*

Study login. The participant creates an account at `sightvr.org` with the study email below and the password from the login card. The study email ties the VR results to the participant ID, not a personal email.

> **SAY** *Please create an account with this email and the password on the login card. We use this login only to connect your training results to your participant number; we do not ask for your personal email.*

SIGHT VR account at `sightvr.org`

Email `p____@sight.study`

Password on the login card

☐ Account created ☐ Login problem (describe on the Machine 1 sheet)

Safety note

Remind the participant they may pause or stop at any time. **Fatigue, pain, dizziness, or loss of balance:** you can pause and/or stop the experiment; note it on the observation sheet. Anyone with VR effects (disorientation, eye strain, headache, nausea, sweating) stays at the lab until they pass.

How much help to give (all machines)

1. Observe silently.
2. Safety or technical support only (e.g., the screen is frozen or the system will not start).
3. If the participant is unable to move forward, one neutral prompt: *"Please continue in the way that feels most natural to you."*
4. If the participant asks for help: *"I am not able to guide you through the system, but you may use any instructions available in the system."*

Log each request on the observation sheet. Do not ask for feedback during training.

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Machine 1 of 3: _____

Timed exploration: 15 minutes. The participant does not have to finish every module.

Start time _____ 15:00 mark _____ Observer _____

☐ Stopped at 15:00 ☐ Finished early at _____ ☐ Participant stopped early (note why)

> **SAY** *Please work through the training modules for the _____. The system will guide you through the necessary tasks within the training. You will have about 15 minutes to work through as much of the training module as you are capable. Do not worry if you do not make it through all components of the training.*

Modules reached

Training module	Completed	Clock time and notes
Module 1	☐	
Module 2	☐	
Module 3	☐	
Module 4	☐	
Module 5	☐	
Module 6	☐	

Help requests and points where the participant got stuck

Time	What happened (type of difficulty)	Help level given
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none

Discomfort: ☐ None ☐ Fatigue ☐ Pain ☐ Dizziness ☐ Loss of balance ☐ Other **Any: pause and/or stop**

Next: 5-minute break (restroom, water, walk around), then Experiment #3, Machine 2.

Observation notes: confusion, frustration, and the participant's comments word for word

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Machine 2 of 3: _____

Timed exploration: 15 minutes. The participant does not have to finish every module.

Start time _____ 15:00 mark _____ Observer _____

☐ Stopped at 15:00 ☐ Finished early at _____ ☐ Participant stopped early (note why)

> **SAY** *Please work through the training modules for the _____. The system will guide you through the necessary tasks within the training. You will have about 15 minutes to work through as much of the training module as you are capable. Do not worry if you do not make it through all components of the training.*

Modules reached

Training module	Completed	Clock time and notes
Module 1	☐	
Module 2	☐	
Module 3	☐	
Module 4	☐	
Module 5	☐	
Module 6	☐	

Help requests and points where the participant got stuck

Time	What happened (type of difficulty)	Help level given
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none

Discomfort: ☐ None ☐ Fatigue ☐ Pain ☐ Dizziness ☐ Loss of balance ☐ Other **Any: pause and/or stop**

Next: 5-minute break (restroom, water, walk around), then Experiment #4, Machine 3.

Observation notes: confusion, frustration, and the participant's comments word for word

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Machine 3 of 3: _____

Timed exploration: 15 minutes. The participant does not have to finish every module.

Start time _____ 15:00 mark _____ Observer _____

☐ Stopped at 15:00 ☐ Finished early at _____ ☐ Participant stopped early (note why)

> **SAY** *Please work through the training modules for the _____. The system will guide you through the necessary tasks within the training. You will have about 15 minutes to work through as much of the training module as you are capable. Do not worry if you do not make it through all components of the training.*

Modules reached

Training module	Completed	Clock time and notes
Module 1	☐	
Module 2	☐	
Module 3	☐	
Module 4	☐	
Module 5	☐	
Module 6	☐	

Help requests and points where the participant got stuck

Time	What happened (type of difficulty)	Help level given
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none

Discomfort: ☐ None ☐ Fatigue ☐ Pain ☐ Dizziness ☐ Loss of balance ☐ Other **Any: pause and/or stop**

Next: 5-minute break (restroom, water, walk around), then End of Session #1, Assessment Modules.

Observation notes: confusion, frustration, and the participant's comments word for word

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Assessment Modules

One fixed assessment module per machine, taken in the order the machines were trained

Every participant takes the same module for each machine: CNC Lathe, Module 4 (Program the CNC Machine); Mill, Module 5 (Configure the Milling Spindle); Cobot, Module 2 (Boot the Cobot Arm).

> **SAY** *Now you will complete one short assessment module for each machine, in the same order you trained on them. Please work through each assessment on your own.*

Order	Machine	Assessment module	Start	End	Completed?	Notes
___	Mill	Module 5: Configure the Milling Spindle			☐ Yes ☐ No	
___	CNC Lathe	Module 4: Program the CNC Machine			☐ Yes ☐ No	
___	Cobot	Module 2: Boot the Cobot Arm			☐ Yes ☐ No	

Notes on the assessments

Next: hand the participant the TAM questionnaire (End of Session #2, in the participant forms).

> **SAY** *Now, please complete this survey. It will ask about your experience using the system. You may leave questions blank that you do not want to answer.*

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Post-Session Interview

Read each question aloud, then write the participant's answer and key quotes

SAY *Now, let's talk a little bit about what made the system easy or difficult to use. If you agree, I will audio record our interview.*

Audio recording

☐ Participant agreed: recording started at _____ ☐ Participant declined: notes only

If audio consent is not given, write the answers on these pages. When recording on the study recorder, still note key points and the recorder time stamp beside them.

The audio choice is on the revised consent form, which is still a draft. Do not record until the IRB approves it.

1. If you have taken any other machine safety training courses, how does the SIGHT VR system compare to those training courses?

2. What did you enjoy most about the training modules that you worked through?

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3. Do you have any suggestions on how to improve the SIGHT system to make it more effective and/or enjoyable to use? This can be related to the interface, the specific training content, or the way that the training content is presented.

4. Are there aspects of the SIGHT system that we should make sure to keep in any future revisions?

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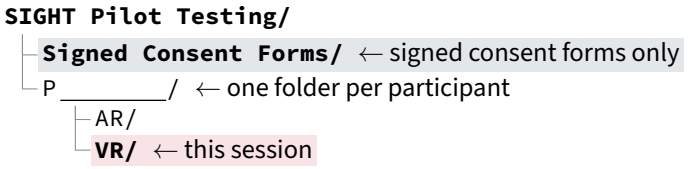
Close-out & Data Entry

Close the session, then upload everything within 24 hours

Close the session

- [ ] Thank the participant for taking part in the study.
- [ ] Check how the participant feels. Anyone with lingering VR effects stays at the lab until they pass.
- [ ] Confirm the date of the AR session, about one week from today: _____
- [ ] Log the study account out of the VR system and close the browser.
- [ ] Give the participant a **$75 Visa gift card** for this session. If they sign a receipt, keep it with the signed consent forms (it includes their name), not in this packet.

Step 1. Archive in Google Drive (shared **SIGHT Pilot Testing** folder): the packet and transcript (never the audio) go in this participant's **VR** subfolder; the signed consent form goes in **Signed Consent Forms**.



- [ ] Scan the participant forms (**without the consent form or the login card**) and this packet together as one PDF and upload it to the **VR** subfolder:
  `P _____VR_packet_YYYY-MM-DD.pdf`
- [ ] Upload the transcript, if recorded: `P _____VR_interview_YYYY-MM-DD_transcript.txt`
- [ ] Scan the signed consent form into `Signed Consent Forms/` as `Consent_YYYY-MM-DD.pdf` (add `_2` for a second one that day). Never name it with the participant ID or put it in a participant folder.
- [ ] Store the paper consent form apart from both packets, in the project's locked storage.

Step 2. Enter the data in the SIGHT Test Hub (`sight-test-hub.vercel.app`; sign in with your Miami Google account)

- [ ] Upload the same packet PDF. If the Hub cannot identify a page, it asks you first.
- [ ] If recorded: transcribe the interview on the SIGHT workstation with the **SIGHT Interview Transcriber**, check it against the recording, then delete the recording from the recorder and the workstation.
- [ ] Upload the transcript file (names as [name]). In **Interview Transcript**, confirm the four answers and tick that the recording is deleted. Then delete the transcript files from the workstation.
- [ ] Open each page, compare every field with the paper, correct anything misread, and confirm the page. Black out any name written on a page.
- [ ] Ask a team member who confirmed none of the pages or the transcript to check the session. Upload the transcript first: one added after the check sends the session back to review.
- [ ] No hand scoring: the Hub scores the JHA quiz, GATT and TAM.